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Estimating core body temperature from heart rate using a residual-compensated adaptive Kalman filter

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ABSTRACT

Accurate, real-time estimation of core body temperature is critical for preventing heat-related illness. While existing Kalman filter-based methods offer interpretable, single-input (heart rate) solutions, they are limited by fixed observation models that fail to capture the complex, non-linear, state-dependent dynamics of physiological signals.

To address this, we propose the Residual-Compensated Adaptive Kalman Filter (RCAKF), a novel hybrid framework. The RCAKF integrates a long short-term memory (LSTM) network to learn and correct structured, state-dependent errors in the observation model, alongside an adaptive noise estimator that dynamically adjusts for measurement uncertainty. This architecture enhances the classic Kalman filter with data-driven flexibility while maintaining its recursive structure and interpretability.

Evaluation was conducted on a controlled experimental dataset with 22 participants performing exercise and recovery under varied thermal conditions. Compared to five baseline models: extended Kalman filter (EKF: RMSE = 0.39 °C), the improved ECTemp model with a sigmoid observation function (ECTemp-S: RMSE = 0.40 °C), biphasic Kalman filter-based model (BKFB: RMSE = 0.48 °C), moving-average Kalman filter (MAKF: RMSE = 0.38 °C), and a standalone LSTM network (RMSE = 0.46 °C), RCAKF achieved the best accuracy with an RMSE of 0.31 °C.

By augmenting the Kalman filter with a learned residual correction and adaptive uncertainty, the RCAKF framework significantly enhances core temperature tracking from a single heart rate signal. Its accuracy and reliance on a single, common sensor make it a practical and promising solution for real-time deployment on wearable devices for safety monitoring.

1. Introduction

Core body temperature is a critical physiological parameter reflecting the thermal homeostasis of the human body. It plays a particularly important role in preventing heat stress-related injuries during strenuous physical activity, occupational labor, or exposure to high ambient temperatures [1–3]. Under such conditions, increased metabolic heat production and reduced heat dissipation can cause core temperature to rise rapidly. If core temperature exceeds 41°C within a short period, it can result in severe thermal injury, including protein denaturation and cellular necrosis. Without timely intervention, further increases in body temperature may lead to heat stroke, a medical emergency with a mortality rate of approximately 12% among adults [4]. According to reports, an average of 5946 individuals are treated for exertional heat-related illness (EHRI) in emergency departments across the United States each year [5]. Therefore, accurate monitoring and early warning of core body temperature during exercise, training, or high-risk work environments is essential for preventing injury and ensuring physiological safety.

Although rectal temperature is often considered too invasive for routine or continuous use, it remains the criterion standard for assessing core temperature in hyperthermic individuals [6,7]. As recommended by the National Athletic Trainers' Association (NATA) position statement, rectal temperature offers valid and reliable measurements both at rest and during exercise, and is considered an appropriate reference method for evaluating core body temperature in physically active settings [8]. In practice, however, peripheral measurement sites-such as the axilla, tympanic membrane, oral cavity, and forehead-are often used for convenience [9,10]. These sites, while more accessible, have been shown to produce substantial mean bias compared to rectal

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temperature, particularly during exertional heat stress. Therefore, they are generally inadequate for real-time or high-precision core temperature monitoring [11,12].

Thermoregulatory models of the human body describe internal heat transfer and physiological control mechanisms, such as vasodilation, vasoconstriction, sweating, and shivering [13,14]. These models are typically grounded in biophysical heat balance equations that account for metabolic heat production, as well as heat exchange with the environment through convection, radiation, and evaporation [15–17]. To represent the spatial heterogeneity of thermal behavior, classical models-such as those developed by Stolwijk and Fiala-divide the human body into multiple anatomical segments and tissue layers, each with its own thermal capacity and blood flow dynamics [18,19]. These node-based models simulate both passive thermal conduction and active regulatory responses, providing detailed predictions of core and skin temperatures under varying environmental and activity conditions.

Despite their physiological interpretability, such models often require extensive individual-specific and environmental inputs-such as metabolic rate, clothing insulation, ambient humidity, and wind speed-which are difficult to measure accurately in real-world settings. Moreover, their deterministic nature may limit adaptability to unmodeled dynamics or inter-individual variability, reducing robustness in uncontrolled or rapidly changing environments [20,21].

With advances in wearable and portable sensing technologies, a variety of methods have been proposed to enable practical and continuous monitoring of core body temperature. Ingestible telemetric capsules provide highly accurate internal readings and are often regarded as a reference standard in field studies, but their invasive nature, high cost, and single-use design limit the practical application [22,23]. Zero-heat-flux thermometry achieves good agreement with esophageal temperature by creating an isothermal tunnel from skin to core, with reported 95 % limits of agreement of $\pm 0.40\,^{\circ}C$ during exercise and minimal time delay, although its performance remains highly dependent on probe placement and heating stability [24,25]. Heat flux sensors, which estimate core temperature from the thermal gradient across dual skin probes, have shown high accuracy in recent studies, achieving a root mean squared error as low as $0.087\,^{\circ}C$ [26]. However, their performance can be influenced by factors such as changes in skin blood flow, sweating and evaporative cooling during exercise [27]. These technologies can achieve excellent accuracy and are valuable in many applications. However, their reliance on dedicated hardware constrains scalability and long-term usability.

In parallel, estimating core temperature directly from readily accessible physiological signals-such as heart rate and skin temperature-has emerged as a more scalable strategy for unobtrusive, real-time monitoring in athletic and occupational settings [28–32]. This approach is particularly valuable in scenarios requiring continuous and unobtrusive monitoring, including athletic performance tracking, heat acclimatization, and occupational safety.

A variety of modeling approaches have been developed based on wearable sensor data, which can be broadly categorized. Firstly, purely data-driven models typically rely on supervised learning algorithms to capture nonlinear mappings between multimodal sensor inputs and core temperature [33]. For example, Niedermann et al. achieved an RMSE of approximately $0.28\,^{\circ}C$ to $0.32\,^{\circ}C$ using multiple linear regression with up to six signals, including heart rate, skin temperature and heat flux [34]. While often accurate, these "black-box" models can require extensive sensor arrays and may lack physiological interpretability.

Secondly, hybrid physics-based models offer a more interpretable alternative by integrating real-time sensor data into a physiological framework. For instance, a recent compartment model proposed by Hirata et al., driven by heart rate and ambient conditions, achieved a high accuracy with an RMSE as low as $0.14\,^{\circ}C$ by using the heart rate data to estimate metabolic rate within a simplified thermodynamic structure [35]. While offering physiological interpretability, these models can still require multiple inputs and carefully tuned parameters.

Thirdly, in contrast to multi-sensor approaches, several studies have explored sequential estimation frameworks that require only minimal physiological inputs. State-space models-particularly Kalman filtering-offer a principled approach by treating core body temperature as a latent state that evolves over time [36,37]. Buller et al. were the first to propose an extended Kalman filter (EKF) for core temperature estimation using heart rate as the sole observation input [37], and their model has since been widely adopted and validated across a range of physical activities and environmental conditions [36,38,39]. By relying exclusively on heart rate, these methods maximize sensor simplicity, enabling practical deployment in wearable devices and field settings where minimizing hardware complexity, energy consumption, and cost is critical. This sensor-efficient design also facilitates applications in large-scale monitoring, athletic training, occupational health, and circadian rhythm analysis. Building upon this foundation, Buller's group further refined their model to improve its accuracy during periods of rest by replacing the original quadratic observation model with a generalized logistic (sigmoid) function [40]. This modification better captured the saturation behavior of heart rate at both low and high core temperatures, thereby reducing systematic underestimation at rest and enhancing circadian rhythm estimation potential.

Recent developments in EKF-based core temperature estimation can be broadly grouped under a paradigm we refer to as Macro-Strategy Switching. This approach enhances model adaptability by employing multiple pre-defined EKF models, each designed for a specific physiological phase. For instance, the method proposed by Rizvi and Udayraj switches models based on a moving average of heart rate trends [30], while the more recent Biphasic Kalman Filter-Based (BKFB) model by Falcone et al. uses a simple heart rate threshold to switch between 'exercise' and 'recovery' filters [41]. While these approaches offer interpretable, computationally efficient solutions that improve upon a single static model, their switching is discrete and reactive. They remain fundamentally incapable of addressing the core challenge of non-stationary conditions: the dynamic, time-varying errors inherent in the observation model itself, which arise from relying on fixed-form equations.

To mitigate these limitations, we explore a complementary paradigm: Micro-Level Real-Time Correction. Hybrid models that integrate Kalman filters with deep learning architectures like LSTM have demonstrated considerable success in various engineering and environmental science domains. For example, they have been employed to enhance the accuracy of streamflow forecasting in hydrology [42], to improve state estimation in vehicle dynamics control [43], and to track non-linear trajectories of underwater targets [44].

Inspired by these successes, our work pioneers the application of such a hybrid framework to the distinct challenge of core body temperature estimation. In this work, we introduce the Residual-Compensated Adaptive Kalman Filter (RCAKF) as a concrete implementation of this paradigm. The core philosophy of this approach is not to alternate between models, but to augment a single EKF with mechanisms that enable continuous adaptation. Specifically, the RCAKF integrates a long short-term memory (LSTM) network to model and compensate for structured, time-varying residuals in the observation process, alongside an adaptive noise estimator that dynamically adjusts measurement uncertainty. This hybrid design retains the interpretability and structure of classical Kalman filtering while leveraging the representational power of deep learning. Operating solely on heart rate input, RCAKF offers enhanced flexibility and improved accuracy under diverse physiological conditions.

This study makes three primary contributions. First, we formalize and validate the micro-level real-time correction paradigm in the context of non-invasive core temperature estimation, demonstrating its advantages over existing macro-strategy switching methods. Second, we present the design of RCAKF, a hybrid filtering framework that successfully integrates LSTM-based residual compensation into the adaptive EKF architecture. Third, we provide comprehensive validation showing that RCAKF significantly outperforms established baselines using only a

Table 1
Summary of participant demographics and experimental settings.

Attribute	Value
Number of participants	22 (12 males / 10 females)
Age range (years)	19-36
Mean age (±SD)	25.3 ± 5.2 years
Mean body mass (±SD)	68.5 ± 10.9 kg
Mean height (±SD)	173.8 ± 6.5 cm
Mean body fat (±SD)	15.7 ± 6.2 %
Clothing conditions	Permeable or impermeable coveralls
Environmental settings	• **Setting 1**: 25°C, 50 % RH, impermeable, no solar • **Setting 2**: 40°C, 20 % RH, permeable, no solar • **Setting 3**: 40°C, 20 % RH, impermeable, no solar • **Setting 4**: 30°C, 35 % RH, permeable, solar (530 $W \cdot m^{-2}$ on back) • **Setting 5**: 40°C, 20 % RH, permeable, solar (530 $W \cdot m^{-2}$ on back) • **Setting 6**: 40°C, 35 % RH, permeable, no solar
Activity protocol	• **Setting 1**: 10 min rest → 40 min treadmill walking → rest (22°C, 50 % RH) until rectal temperature drops by 0.4°C → 40 min treadmill walking • **Setting 6**: 10 min rest in climatic chamber → 60 min treadmill walking → rest (22°C, 50 % RH) until 0.4°C drop in rectal temperature → 40 min treadmill walking • **Setting 2–5**: Same protocol as Setting 1
Recorded signals	Heart rate, rectal temperature, activity labels

single sensor input-highlighting its practical value for real-world wearable applications.

2. Methods

2.1. Dataset

This study utilized a specific subset of physiological data from the PROSPIE project [45–47]. The selected dataset included six distinct experimental settings designed to simulate varying levels of thermal strain in healthy adult participants. These settings differed in ambient temperature (ranging from 25 °C to 40 °C), relative humidity (20 % to 50 %), and clothing ensembles (permeable cotton versus impermeable nylon coveralls). Additionally, some sessions included a simulated solar load of 500 $W \cdot m^{-2}$, while others were conducted without solar radiation.

Each session generally followed a multi-phase structure: (1) an initial seated rest, (2) a treadmill walking phase (typically 40 minutes, but extended to 60 minutes in one condition), (3) a standardized recovery phase, and (4) a second walking phase.

Throughout each session, heart rate (HR), rectal temperature (used as the reference for core temperature), and activity stage labels were continuously recorded. Participants with missing measurement values were excluded from the analysis to ensure model robustness. The final six settings included 12, 16, 18, 15, 19, and 15 complete trials, respectively. A summary of the specific conditions and participant demographics is provided in Table 1.

2.2. Extended Kalman Filter (EKF)

To enable non-invasive estimation of core body temperature from heart rate, this study employs an extended Kalman filter (EKF) framework in which core temperature is treated as a latent state and heart rate as the observed signal [38,39]. The EKF recursively updates the temperature estimate by combining a dynamic time update model and a nonlinear observation model, both empirically derived from training data.

The *time update model* describes the evolution of core temperature over time and is implemented as a first-order autoregressive process learned via linear regression. This model captures the gradual physiological change in temperature and introduces process noise to reflect uncertainty.

The *observation model* establishes the mapping from core temperature to heart rate using a second-order polynomial regression, reflecting their nonlinear physiological relationship. Observation noise is modeled to account for deviations between predicted and actual heart rate.

Together, these models allow the EKF to balance prior knowledge with real-time heart rate measurements, enabling sequential, real-time estimation of core body temperature from wearable signals.

State transition model.

$$T_t = a_1 T_{t-1} + a_0 + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \gamma^2) \tag{1}$$

Here, T_t is the latent core temperature at time t, and T_{t-1} is the previous estimate. The parameters a_1 and a_0 are obtained from linear regression, and γ^2 represents the process noise variance.

Observation model.

$$\hat{HR}_t = b_2 T_t^2 + b_1 T_t + b_0 + \eta_t, \eta_t \sim \mathcal{N}(0, \sigma^2) \tag{2}$$

In this model, $\hat{HR}_t$ is the predicted heart rate corresponding to core temperature T_t. The coefficients b_2, b_1, and b_0 are obtained from second-order polynomial regression. The term η_t is Gaussian observation noise with variance σ^2.

The recursive filtering process proceeds as follows:

Prediction step.

$$\hat{T}_t = a_1 T_{t-1} + a_0 \tag{3}$$

$$\hat{v}_t = v_{t-1} + \gamma^2 \tag{4}$$

The prior temperature estimate $\hat{T}_t$ and its associated variance $\hat{v}_t$ are predicted using the state transition model.

Observation prediction.

$$\hat{HR}_t = b_2 \hat{T}_t^2 + b_1 \hat{T}_t + b_0 \tag{5}$$

The expected heart rate is computed from the predicted core temperature.

Extended Kalman filter mapping function variance coefficient.

$$m_t = \frac{d\hat{HR}_t}{dT} = 2b_2 \hat{T}_t + b_1 \tag{6}$$

To enable Kalman-based updates with a nonlinear observation model, the function is linearized and the term m_t denotes the extended Kalman filter mapping function variance coefficient.

Kalman gain.

$$k_t = \frac{\hat{v}_t \cdot m_t}{m_t^2 \cdot \hat{v}_t + \sigma^2} \tag{7}$$

The Kalman gain k_t determines the weight given to the new observation in updating the state estimate.

Update step.

$$T_t = \hat{T}_t + k_t \cdot (HR_t - \hat{HR}_t) \tag{8}$$

The posterior temperature estimate is corrected based on the observed heart rate HR_t.

Variance update.

$$v_t = (1 - k_t m_t) \cdot \hat{v}_t \tag{9}$$

The variance is updated to reflect the new estimation uncertainty.

The EKF is initialized with a core temperature of $T_0 = 37.0°C$ and a variance of $v_0 = 0.01\ °C^2$ in this study. All model parameters are fitted from training data and fixed during inference. This filter offers a lightweight and interpretable solution for real-time estimation of core body temperature using heart rate data alone.

2.3. Residual-Compensated Adaptive Kalman Filter (RCAKF)

To improve the robustness of core temperature estimation in the presence of time-varying and non-Gaussian noise, we propose a Residual-Compensated Adaptive Kalman Filter (RCAKF). This method extends the standard extended Kalman filter (EKF) by incorporating a deep learning-based residual compensation module and an adaptive observation noise estimator.

Specifically, the RCAKF augments the polynomial observation model with a data-driven correction term predicted by a long short-term memory (LSTM) network. Additionally, the observation noise variance is no longer assumed fixed but is dynamically estimated from recent prediction errors via a sliding window approach.

Observation model with residual correction. To account for systematic, state-dependent deviations in the predicted heart rate not captured by the polynomial model, we introduce a learned residual correction term $\hat{r}_t$, estimated by an LSTM network:

$$\hat{r}_t = \mathrm{LSTM}(HR_{t-n:t-1}) \tag{10}$$

The corrected observation becomes:

$$\hat{HR}_t = b_2 \hat{T}_t^2 + b_1 \hat{T}_t + b_0 + \hat{r}_t \tag{11}$$

This addition allows the observation model to dynamically adapt its output based on recent heart rate patterns, which is crucial for handling non-stationary physiological states.

Prediction error and adaptive noise estimation. The prediction error at time t is computed as:

$$\epsilon_t = HR_t - \hat{HR}_t \tag{12}$$

Furthermore, to account for time-varying measurement uncertainty, we replace the fixed observation noise assumption by adaptively estimating the variance $\hat{\sigma}_t^2$ from a sliding window of recent prediction errors, where a buffer $S = \{\epsilon_{t-w+1}^2, \ldots, \epsilon_t^2\}$ of length w is maintained and used for variance estimation:

$$\hat{\sigma}_t^2 = \max\left(0, \mathrm{mean}(S) - m_t^2 \hat{v}_t\right) \tag{13}$$

where $m_t = \frac{d\hat{HR}_t}{dT} = 2b_2\hat{T}_t + b_1$ is the linearized observation model derivative, and $\hat{v}_t$ is the predicted state variance.

Kalman gain and state update. The Kalman gain is computed using the adapted noise variance:

$$k_t = \frac{\hat{v}_t \cdot m_t}{m_t^2 \cdot \hat{v}_t + \hat{\sigma}_t^2} \tag{14}$$

$$T_t = \hat{T}_t + k_t \cdot \epsilon_t \tag{15}$$

$$v_t = (1 - k_t m_t) \cdot \hat{v}_t \tag{16}$$

The main steps of the proposed RCAKF method are summarized in Algorithm 1.

Algorithm 1 Residual-Compensated Adaptive Kalman Filter (RCAKF).

1. **Input:** Initial core temperature T_0, variance v_0;
   coefficients a_0, a_1, b_0, b_1, b_2;
   process noise std γ; initial observation noise std σ_0;
   residual model LSTM; LSTM window size n; adaptive window size w
2. Initialize squared error buffer: $S \leftarrow [\sigma_0^2] \times w$
3. For each time step $t = 1$ to T, do:

   Step 1: Predict core temperature and its variance:

   $$\hat{T}_t = a_1 T_{t-1} + a_0, \quad \hat{v}_t = v_{t-1} + \gamma^2$$

   Step 2: Estimate residual using LSTM:

   $$\hat{r}_t = \mathrm{LSTM}(HR_{t-n:t-1})$$

   Step 3: Observation prediction with residual compensation:

   $$\hat{HR}_t = b_2\hat{T}_t^2 + b_1\hat{T}_t + b_0 + \hat{r}_t, \quad m_t = 2b_2\hat{T}_t + b_1$$

   Step 4: Compute prediction error and update observation noise estimate:

   $$\epsilon_t = HR_t - \hat{HR}_t$$

   $$S \leftarrow S[1:] \cup \{\epsilon_t^2\}, \quad \hat{\sigma}_t^2 = \max(0, \mathrm{mean}(S) - m_t^2 \hat{v}_t)$$

   Step 5: Compute Kalman gain and update:

   $$k_t = \frac{\hat{v}_t m_t}{m_t^2 \hat{v}_t + \hat{\sigma}_t^2}$$

   $$T_t = \hat{T}_t + k_t \cdot \epsilon_t, \quad v_t = (1 - k_t m_t) \cdot \hat{v}_t$$

4. **Output:** Estimated temperature sequence $\{T_t\}_{t=1}^{T}$

Implementation details. To ensure compatibility with wearable deployment constraints, the residual model is intentionally designed to be lightweight. It consists of a single-layer LSTM network with a hidden dimension of 32, trained using the Adam optimizer with a learning rate of 0.001. Early stopping is applied with a patience of 5 epochs based on validation loss.

The model operates solely on a univariate input sequence of heart rate measurements, eliminating the need for multimodal signals or demographic information. Specifically, the LSTM takes as input a sliding window of the past $n = 10$ heart rate values. Despite the input simplicity, this architecture is sufficient to capture structured deviations in the observation model under varying physiological conditions.

To evaluate model generalization across sessions and experimental conditions, we adopt a nested 3-fold cross-validation strategy based on trial-level splitting. Each trial corresponds to an independent data collection session performed under specific thermal, activity, and clothing conditions. Trials are randomly divided into three disjoint folds, ensuring complete separation between training and testing trials. In each

outer fold, an 80/20 split of the training set is used to construct an inner validation set for early stopping and adaptive parameter tuning. All model parameters-including autoregressive coefficients, polynomial observation mappings, and LSTM weights-are fitted exclusively using the outer training trials.

The adaptive variance window size w is selected by grid search over $\{5, 10, 20, 30, 50\}$ using inner validation performance. This configuration ensures that the residual-compensated adaptive Kalman filter (RCAKF) remains computationally efficient and deployable in real time on wearable platforms, while maintaining robust performance across unseen trial conditions.

2.4. Evaluation metrics

To assess model performance and generalization, we employed a nested 3-fold cross-validation procedure based on complete experimental trials. Each outer fold contained a disjoint subset of entire trials, ensuring that no partial session was used simultaneously for training and testing. This validation strategy evaluates the model's ability to generalize across different thermal environments, clothing conditions, and physical activity patterns.

We assessed model accuracy using three standard metrics: root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). These metrics were computed between the estimated and reference core temperature values across all time points in the test sets:

- **RMSE** quantifies the standard deviation of estimation errors and penalizes large deviations:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(T_t - \hat{T}_t)^2}$$

- **MAE** measures the average magnitude of absolute errors:

$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N}|T_t - \hat{T}_t|$$

- **Coefficient of Determination** (R^2) reflects the proportion of variance in the reference values explained by the estimates:

$$R^2 = 1 - \frac{\sum_{t=1}^{N}(T_t - \hat{T}_t)^2}{\sum_{t=1}^{N}(T_t - \bar{T})^2}$$

where $\bar{T}$ denotes the mean of the reference core temperature.

In addition to standard error metrics, we employed Bland-Altman analysis to evaluate the agreement between estimated and reference core temperatures [48]. This method plots the difference $(\hat{T}_t - T_t)$ against the mean $(\hat{T}_t + T_t)/2$, providing insight into systematic over- or underestimation across the temperature range.

The analysis reports the mean difference (MD), representing the average bias, and the standard deviation (SD) of the differences. The 95 % limits of agreement, calculated as $\mathrm{MD} \pm 1.96 \cdot \mathrm{SD}$, indicate the interval within which most residuals are expected to fall under the assumption of normality [49]. This graphical approach complements numerical metrics by revealing trends and potential heteroscedasticity in the prediction errors.

To further assess each model's ability to identify high-risk core temperature episodes, we conducted receiver operating characteristic (ROC) analysis [50]. The threshold of 38.0°C was adopted, consistent with occupational heat stress screening guidelines [51]. Binary classification performance was quantified using the area under the ROC curve (AUC), sensitivity (true positive rate), specificity (true negative rate).

These threshold-based metrics offer practical insights into the models' applicability in early warning systems and real-time decision-making, under both conservative and high-risk definitions of elevated core temperature.

2.5. Comparison methods

To evaluate the effectiveness of the proposed residual-compensated adaptive Kalman filter (RCAKF), we compared its performance with several representative baselines encompassing both classical filtering and learning-based approaches:

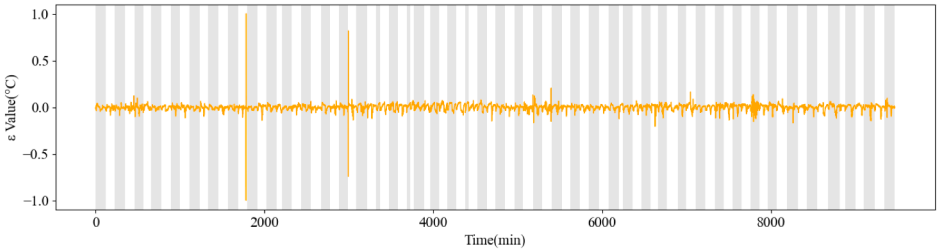
- **Extended Kalman Filter (EKF)**: This baseline implements a standard extended Kalman filter using a single global state transition and observation model, learned from all training data. The observation model is a second-order polynomial mapping from core temperature to heart rate, and the filter does not adapt to state-dependent or time-varying residual patterns.
- **Improved ECTemp™(ECTemp-S)**: Based on the refined ECTemp™ algorithm by Looney and Buller [40], this method replaces the quadratic observation model with a generalized logistic (sigmoid) function that better captures the saturation behavior of heart rate across the full temperature range. The sigmoid model parameters are fit to the training data via non-linear regression, and its derivative is used in the Kalman update step. This approach enhances estimation accuracy during resting periods while maintaining fidelity during exercise.
- **Biphasic Kalman Filter-Based Model (BKFB)**: Adapted from Falcone et al. [41], this method employs two separate EKF models for the *exercise* and *rest* phases. Phase switching is performed using a heart rate threshold-based rule: if three consecutive heart rate values exceed a predefined threshold, the exercise-phase model is applied; otherwise, the rest-phase model is used. The optimal threshold is selected from the range 100–120 bpm via cross-validation.
- **Moving Average Kalman Filter (MAKF)**: This method was proposed by Rizvi and Udayraj [30] to improve temperature estimation across different physiological phases. It uses two separate extended Kalman filters: one for the *exercise* phase and another for the *recovery* phase, each with its own transition and observation models.

  The active model is determined by the trend of a five-minute moving average of heart rate. Specifically, if the current moving average is lower than the previous one, the recovery-phase model is selected; otherwise, the exercise-phase model is used. This switching rule allows the filter to adapt to changes in cardiovascular response during physical activity and rest.
- **LSTM**: A deep learning model composed of a single-layer long short-term memory (LSTM) network with a hidden dimension of 32. The model receives a 10-minute window of past heart rate values as input and directly outputs the predicted core temperature at the current time step.

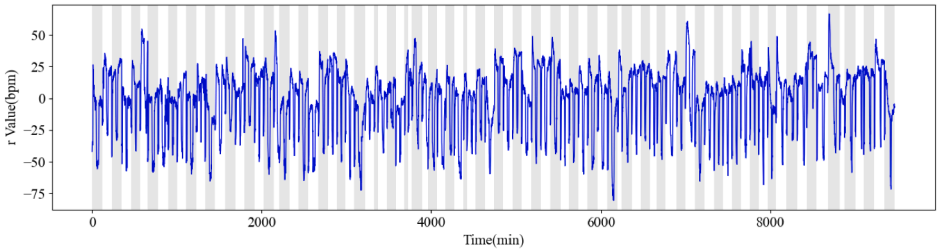
The network architecture and input format are identical to those used in the RCAKF model; the only difference lies in the prediction target. Whereas RCAKF uses the LSTM to estimate observation residuals, the LSTM model here directly regresses the core temperature. This provides a non-recursive, data-driven reference.

All methods are evaluated using a nested 3-fold cross-validation procedure with disjoint experimental trials. In each outer fold, an 80/20 split of the training set is used to construct an inner validation set for early stopping and adaptive parameter tuning. In each outer fold, the dataset is split into training and testing trials to assess generalization. Within the training folds, an inner 3-fold cross-validation is performed to select model-specific hyperparameters, such as the heart rate threshold in BKFB and the adaptive window size in RCAKF. All baseline models were retrained from scratch on the same dataset following the same pipeline, thereby ensuring a fair and direct comparison.

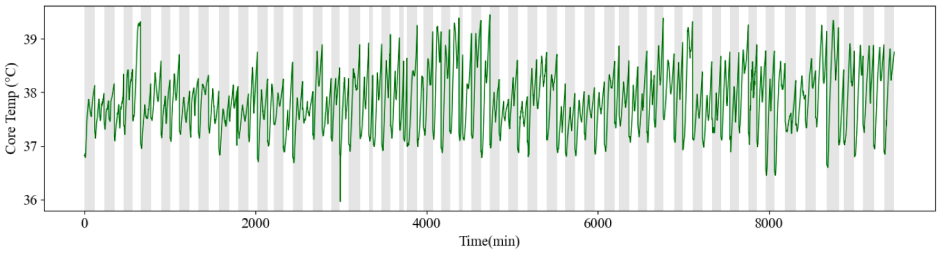
Model parameters are trained exclusively on the outer training folds, and all performance metrics are reported based on the outer test folds to ensure an unbiased and consistent comparison.



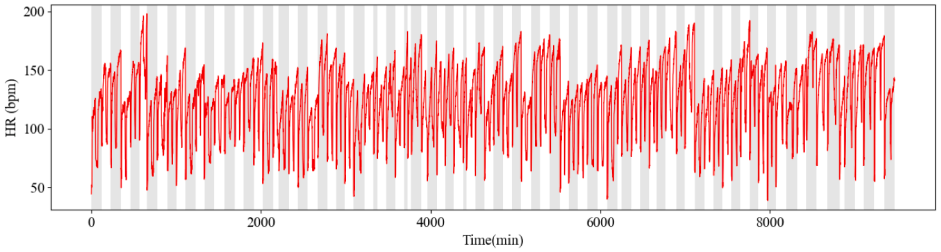
(a) State transition residuals (ϵ_t)



(b) Observation residuals (r_t)



(c) Core temperature



(d) Heart rate (HR)

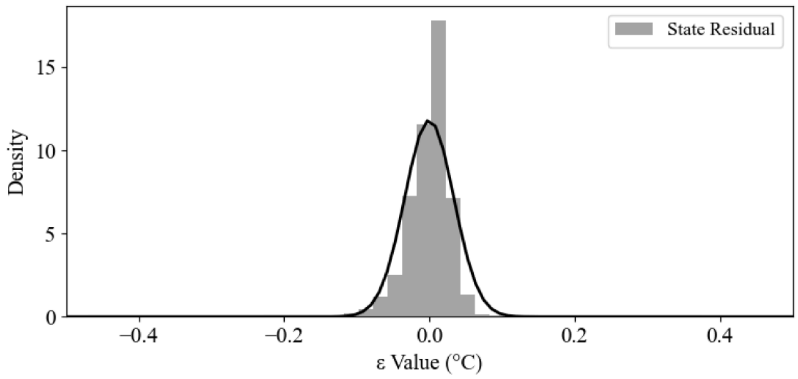
Fig. 1. Overview of residuals and physiological signals across all test trials. Grey and white background segments differentiate trials. A vertical dashed line indicates the boundary between permeable and impermeable clothing conditions.

3. Results

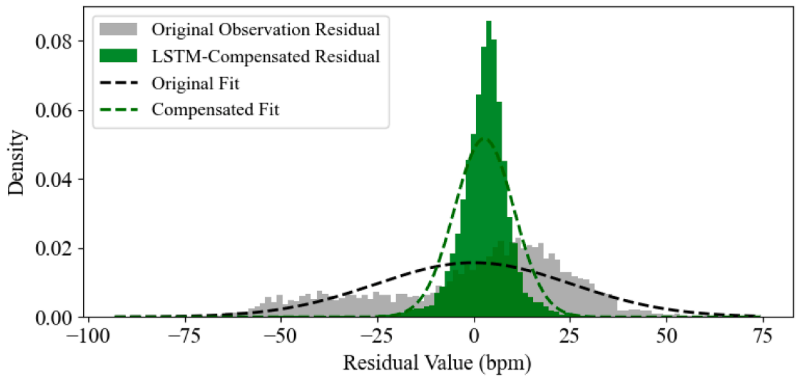
Fig. 1 presents an overview of the physiological time series and residual patterns across all participants and experimental sessions. All time series are concatenated across trials, with alternating grey and white background shades used to differentiate trials. Subplot (a) shows the residuals of the state transition model, computed as the difference between the observed core temperature and its predicted value from the regression model. Subplot (b) presents the observation model residuals, defined as the difference between the measured heart rate and the predicted value from the second-order polynomial regression based on core temperature. Subplots (c) and (d) display the raw core temperature and heart rate trajectories, respectively.

Fig. 2 illustrates the distribution of residuals from the state transition model and the observation model across all participants in the test set. Fig. 2(a) shows the histogram of the prediction error $\epsilon_t = T_t - \hat{T}_t$ from the state transition model, which approximates a zero-mean Gaussian distribution. Fig. 2(b) presents the distribution of observation residuals $r_t = HR_t - H\hat{R}_t$, comparing the fixed-form polynomial observation model with the LSTM-compensated version used in RCAKF. The residuals from the original observation model exhibit substantial skewness, deviating from Gaussianity. In contrast, the residuals after LSTM compensation are approximately zero-mean and symmetric, indicating that the residual correction mechanism effectively accounts for systematic, time-varying deviations in the observation process.

Fig. 3 illustrates the relationship between core temperature and heart rate under different physiological states. The Before LSTM Compensation subplot shows the original mapping between core temperature and observed heart rate, revealing clear state-dependent deviations: heart



(a) State transition residuals (ϵ_t)



(b) Observation residuals (r_t)

Fig. 2. Histogram of model residuals across the test set: (a) Residuals from the state transition model; (b) Residuals from the observation model.

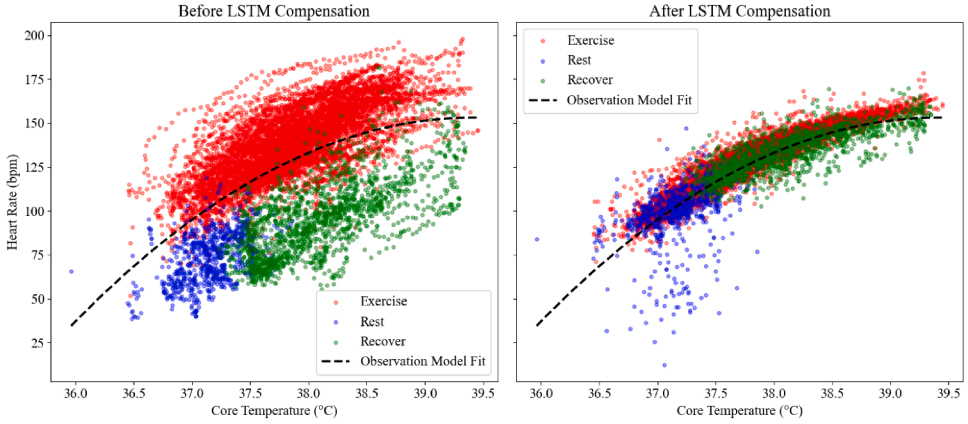


Fig. 3. State-dependent relationship between core temperature and heart rate before and after LSTM-based residual compensation. Left: Original mapping between core temperature and observed heart rate, showing state-specific deviations. Right: Compensated mapping after adding LSTM-predicted residuals to the model output. Each point is color-coded by physiological state Rest, Recovery, or Exercise.

rate tends to be underestimated during *Exercise* and overestimated during *Rest* and *Recovery*. These structured residuals reflect the limitations of using a fixed-form observation model across varying physiological conditions.

The After Compensation subplot presents the relationship after applying LSTM-based residual correction, where the compensated heart rate is computed by adding the predicted residual to the original model output. The corrected mapping exhibits improved alignment across states, with reduced bias and variance. This demonstrates that the RCAKF framework effectively accounts for state-dependent deviations by incorporating a data-driven compensation mechanism.

Taken together, Figs. 1–3 reveal critical limitations of the fixed-form observation model in accurately capturing the dynamics of heart rate responses under varying physiological states. While the state transition residuals largely conform to Gaussian assumptions, the observation residuals show systematic, state-dependent deviations-particularly during *Rest* and *Recovery* phases. These patterns underscore the need for a more flexible observation model capable of learning complex, temporal dependencies. The RCAKF framework addresses this challenge

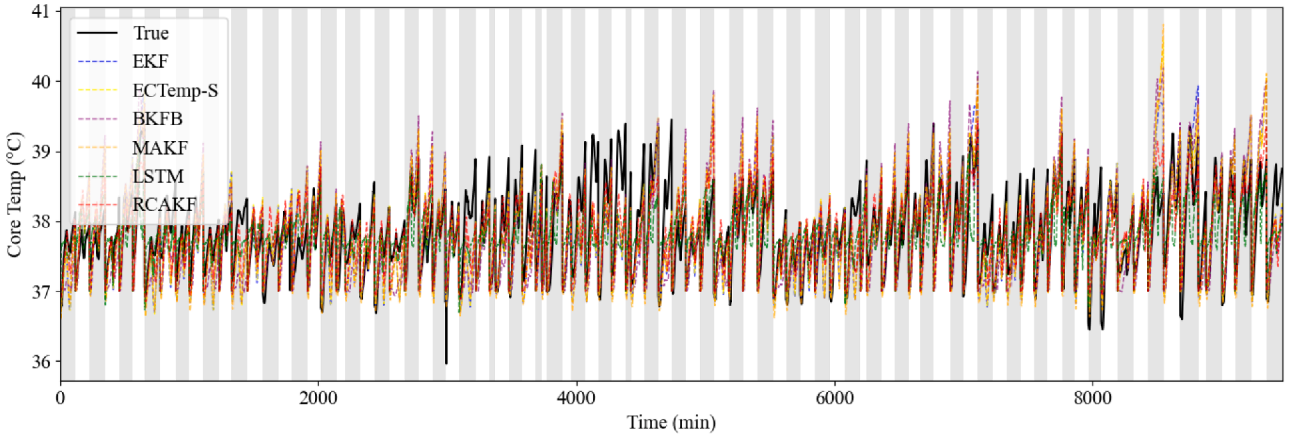


Fig. 4. Time-series comparison of predicted and reference core temperatures across all test trials. The black curve represents the reference core temperature, while colored curves correspond to different models, including EKF, ECTemp-S, BKFB, MAKF, LSTM, and the proposed RCAKF. Alternating grey and white background regions separate individual trials for visual clarity.

by augmenting the standard EKF with a learned residual correction module, enabling improved core temperature estimation performance across diverse physiological conditions.

Fig. 4 presents time-series plots comparing the estimated core temperatures from all models against the ground truth across the entire test set. Each colored line represents the prediction of a specific method, including EKF, ECTemp-S, BKFB, MAKF, LSTM, and the proposed RCAKF. The black line denotes the reference core temperature. Trials are concatenated along the time axis, and alternating grey and white background regions are used to visually separate individual sessions. This figure provides a holistic view of each model's ability to track temperature dynamics across diverse experimental conditions.

Fig. 5 presents Bland–Altman plots comparing the predicted and reference core temperatures. Each plot shows the mean difference (MD) and 95 % limits of agreement (MD ± 1.96 SD), highlighting the magnitude and consistency of model biases. The RCAKF method demonstrates the narrowest spread and the smallest mean bias, indicating superior agreement with the ground truth across the full temperature range.

Fig. 6 shows normalized histograms of estimation errors ($\hat{T}_t - T_t$). The RCAKF distribution is narrower and more centered, indicating improved accuracy and reduced bias.

Fig. 7 presents the receiver operating characteristic (ROC) curves for all six models under the 38.0°C threshold. The ROC curves illustrate each model's ability to distinguish between episodes of normal and elevated core temperature (≥38.0°C), with the area under the curve (AUC) quantifying their binary classification performance.

Table 2 summarizes the comprehensive evaluation results for all six methods across the full dataset. The upper portion reports standard regression metrics, including root mean squared error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and mean difference ± standard deviation (MD ± SD) from Bland–Altman analysis. The lower portion presents classification metrics for detecting elevated core temperature episodes using a fixed threshold of 38.0°C, including the area under the ROC curve (AUC), sensitivity, specificity, and overall accuracy.

Across most of metrics, the proposed RCAKF model outperforms the baseline methods. It achieves the lowest RMSE(0.31 °C) and MAE(0.23 °C), the highest R^2(0.67), and the smallest standard deviation(-0.05 ± 0.31 °C). RCAKF also shows superior ability to identify high-risk core temperature episodes, with the highest AUC (0.9038) and sensitivity (0.8540) confirming its reliability in safety-critical classification performance.

Table 2
Comprehensive evaluation metrics for all models on the full dataset.

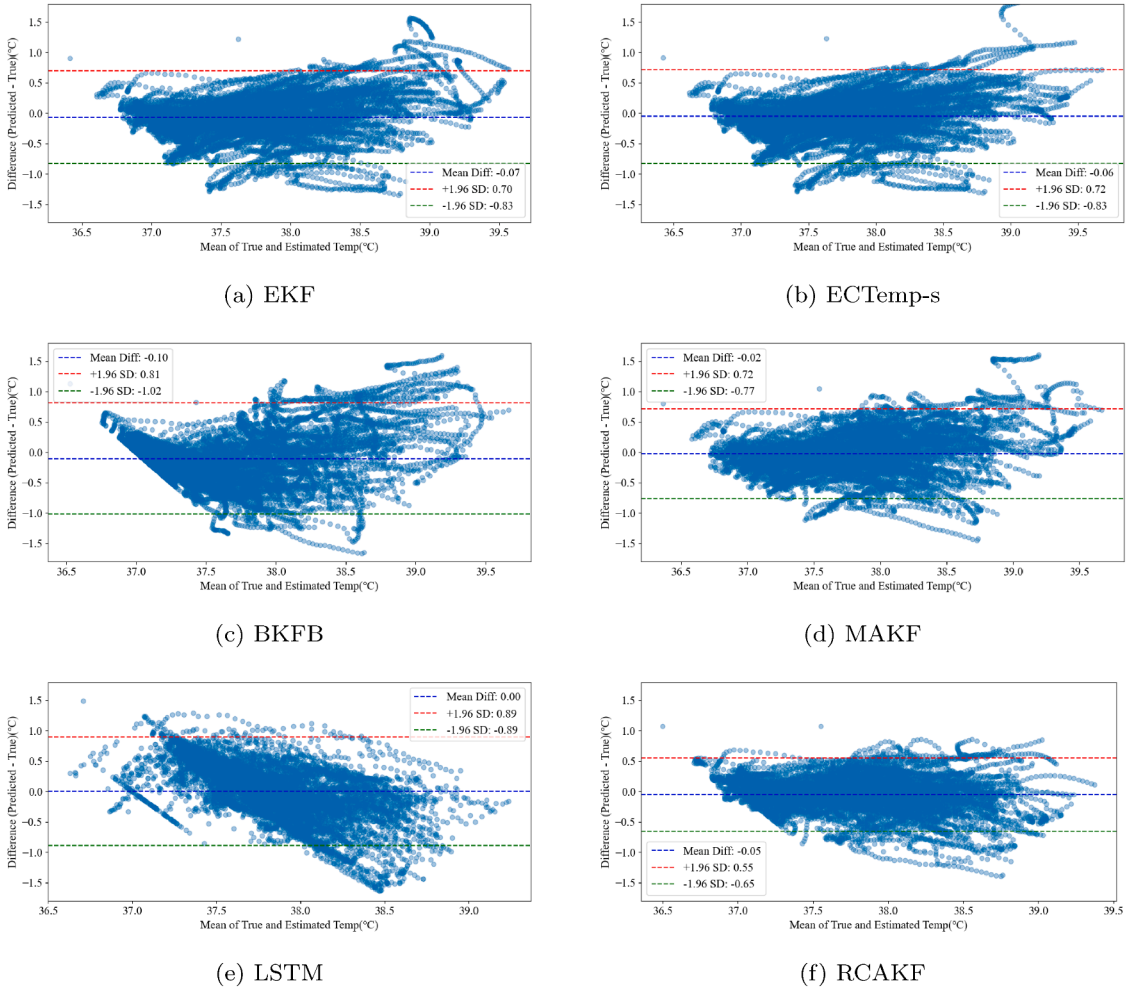
Method	RMSE (°C)	MAE (°C)	R^2	MD ± SD (°C)
EKF	0.39	0.30	0.48	−0.07 ± 0.39
ECTemp-S	0.40	0.29	0.47	−0.06 ± 0.39
BKFB	0.48	0.37	0.24	−0.10 ± 0.47
MAKF	0.38	0.29	0.52	−0.02 ± 0.38
LSTM	0.45	0.35	0.31	0.00 ± 0.45
RCAKF	**0.31**	**0.23**	**0.67**	**-0.05 ± 0.31**
Method	**AUC**	**Sensitivity**	**Specificity**	
EKF	0.8949	0.8116	**0.8279**	
ECTemp-S	0.8950	0.8188	0.8221	
BKFB	0.8858	0.8180	0.7852	
MAKF	0.8963	0.8380	0.7992	
LSTM	0.7563	0.5812	0.8414	
RCAKF	**0.9038**	**0.8540**	0.7978	

4. Discussion

This study proposes a Residual-Compensated Adaptive Kalman Filter (RCAKF) for continuous core temperature estimation using heart rate as the sole input signal. Unlike traditional filtering methods that rely on fixed-form observation models, the RCAKF integrates a lightweight LSTM module to dynamically compensate for structured residuals in the heart rate-temperature relationship. In addition, it incorporates an adaptive observation noise estimation mechanism to account for non-stationary physiological responses across diverse thermal and activity conditions. Together, these enhancements enable more robust and personalized temperature tracking across unseen experimental trials.

Visualization of the heart rate-core temperature mapping before and after residual compensation (Fig. 3) reveals systematic, state-dependent deviations in the observation model. Although the residuals $r_t = HR_t - \hat{H}R_t$ from the uncompensated EKF model are approximately zero-mean overall, their distribution deviates markedly from the Gaussian assumption. During *Exercise*, heart rate is consistently underestimated, while during *Rest* and *Recovery*, it is often overestimated. This asymmetric pattern violates the EKF's assumption that observation noise is independent, identically distributed, and Gaussian.

These deviations not only challenge the statistical assumptions underpinning the EKF, but also degrade its ability to track physiological dynamics under varying conditions. The fixed-form observation model lacks the flexibility to capture the nonlinear, state-specific relationship between heart rate and core temperature, highlighting the need for a more adaptive approach. Crucially, the structured, time-varying nature



(a) EKF (b) ECTemp-s

(c) BKFB (d) MAKF

(e) LSTM (f) RCAKF

Fig. 5. Bland–Altman plots comparing estimated and reference core temperatures for the six models: (a) EKF; (b) ECTemp-S; (c) BKFB; (d) MAKF; (e) LSTM; (f) RCAKF. Each plot shows the mean difference (blue line) and the 95 % limits of agreement (green and red dashed lines) between predicted and ground-truth temperatures.

of these residuals demands a correction mechanism capable of modeling temporal dependencies. Long Short-Term Memory (LSTM) networks are particularly well-suited for this task, as their gated architecture enables the modeling of both short- and long-term temporal dependencies [52]. This capability makes them highly effective in capturing the dynamic and state-dependent patterns characteristic of physiological time series such as heart rate.

To address these limitations, the proposed RCAKF introduces two key innovations to enhance robustness and adaptability. First, an LSTM-based residual correction module learns time-dependent, state-specific deviations not captured by the static polynomial observation model. This correction is applied additively to the predicted heart rate, enabling the filter to account for structured residual patterns that would otherwise bias estimation. As a result, the corrected observation values exhibit more symmetric, zero-centered residual distributions across all physiological states, helping to restore compatibility with the Gaussian noise assumption required by the Kalman framework. Second, RCAKF incorporates an adaptive noise estimation mechanism that dynamically updates the observation noise variance using a sliding window of recent prediction errors. This allows the filter to track fluctuations in uncertainty over time, particularly during transitions between physiological states or under measurement perturbations.

Importantly, these enhancements enable RCAKF to maintain accurate and stable performance under nonstationary conditions, while preserving the interpretability and recursive structure of classical Kalman filters. The latent core temperature state remains explicitly estimated by the Kalman filter, and the neural network's influence is strictly localized to predicting additive corrections for the observation model. To avoid the complexity and instability associated with differentiable Kalman filter formulations, the LSTM module is trained separately using supervised learning to minimize the error between observed and predicted heart rate. During inference, its output is used as a fixed additive term in the observation equation.

This design choice means that the entire RCAKF framework is not trained in an end-to-end manner: gradients do not propagate through the Kalman filter equations, nor is the filter unrolled during training. Instead, the filter operates in a standard recursive manner, using the output of the pre-trained LSTM as a plug-in correction term. This sidesteps the need for implementing differentiable Kalman updates-often a non-trivial task-and ensures numerical stability, implementation simplicity, and clearer interpretability of each module's role. We highlight this to avoid confusion about the training pipeline and to emphasize that RCAKF remains a hybrid system with cleanly decoupled training and inference phases.

Notably, the LSTM module was trained using only heart rate, without requiring contextual labels or environmental features. This minimalist input design enhances the model's transferability to unseen scenarios

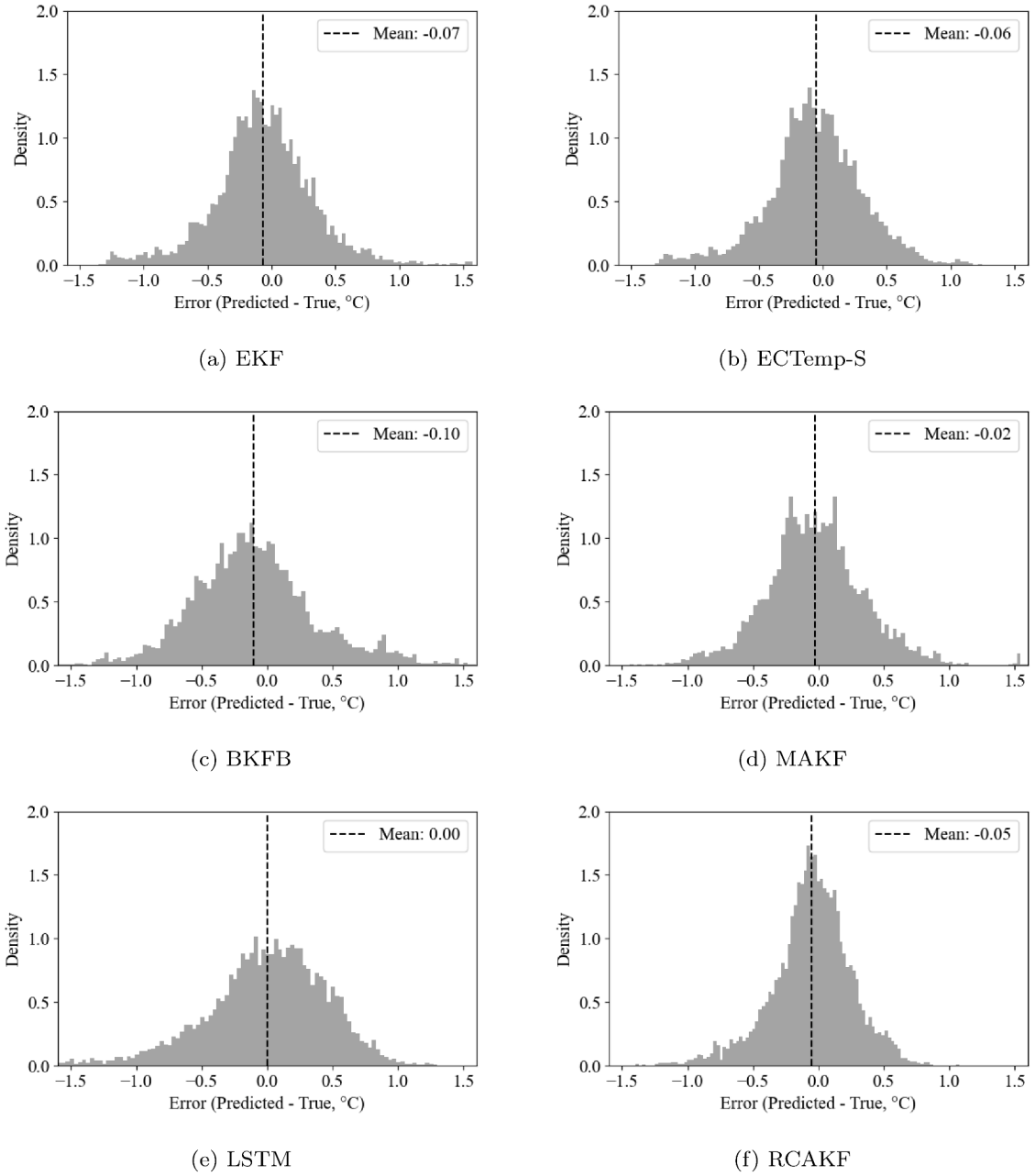


Fig. 6. Normalized histograms of core temperature estimation errors ($\hat{T}_t - T_t$) for six models: (a) EKF; (b) ECTemp-S; (c) BKFB; (d) MAKF; (e) LSTM; (f) RCAKF. All histograms are plotted using the same bin size and range for visual comparability.

and reduces reliance on high-dimensional sensing data, which is advantageous for real-world deployment in wearable systems.

To evaluate the performance of the proposed RCAKF, we compared it against five representative models: the standard extended Kalman filter (EKF), the improved ECTemp model with a sigmoid observation function (ECTemp-S), the biphasic Kalman filter-based model (BKFB), the moving average Kalman filter (MAKF), and a standalone LSTM network.

The EKF serves as a standard and widely used approach for physiological state estimation [39]. It applies fixed-form state transition and observation models and assumes observation noise to be i.i.d. Gaussian. While it achieves moderate performance (RMSE = 0.39, R^2 = 0.48, AUC = 0.895), it cannot account for structured, state-dependent residuals and thus lacks adaptability under dynamic physiological conditions.

The ECTemp-S model modifies the standard EKF by replacing its quadratic observation function with a generalized logistic (sigmoid) mapping from core temperature to heart rate. This adjustment is intended to better reflect the saturating relationship between heart rate and temperature at physiological extremes. In practice, however, ECTemp-S yields performance that is largely comparable to EKF across regression and classification metrics (RMSE = 0.40, R^2 = 0.47, AUC = 0.895). Slight improvements in sensitivity (0.819) and marginally reduced bias (MD ± SD = -0.06 ± 0.39°C) were observed, suggesting that while the sigmoid formulation aligns better with physiological trends, its practical impact under the tested conditions is limited.

The MAKF model-introduced under this name in the current study-extends the EKF by incorporating a phase-switching mechanism based on moving-average heart rate trends. This enables it to differentiate between *exercise* and *recovery* phases, resulting in comparable regression performance but slightly improved classification accuracy over EKF

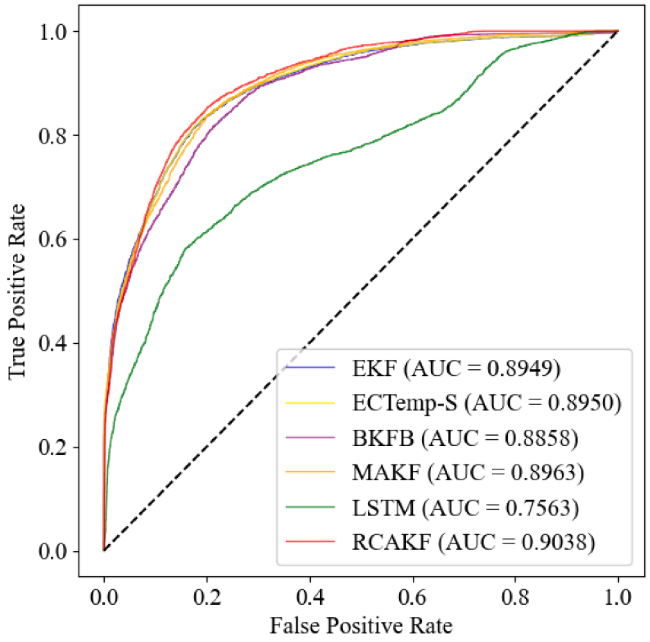


Fig. 7. ROC curves for all six models (EKF, BKFB, MAKF, LSTM, and RCAKF) in detecting elevated core temperature episodes (≥ 38.0°C). The area under the curve (AUC) indicates the discriminative performance of each method.

(RMSE = 0.38, R^2 = 0.52, AUC = 0.896), while maintaining a relatively low bias (MD ± SD = -0.02 ± 0.38°C).

The BKFB method, adapted from Falcone et al. [41], segments the estimation into two fixed filters triggered by a threshold on heart rate. Although conceptually simple, this binary switching strategy leads to degraded performance across both regression and classification metrics (RMSE = 0.48, R^2 = 0.37, AUC = 0.886), indicating limited generalizability.

The standalone LSTM model, which directly regresses core temperature from a 10-minute heart rate window, yields the weakest performance overall (RMSE = 0.45, R^2 = 0.31), with low sensitivity (0.581) despite moderate specificity (0.841). Its purely data-driven design lacks the recursive estimation and uncertainty modeling capabilities of Kalman-based methods.

In contrast, RCAKF consistently outperforms all baselines across almost all of the evaluation metrics. It achieves the lowest error (RMSE = 0.31, MAE = 0.23), highest explanatory power (R^2 = 0.67), and best residual distribution (MD ± SD = -0.05 ± 0.31 °C). In the classification task at the 38.0 °C threshold, it maintains strong discriminative performance (AUC = 0.904) and the highest sensitivity (0.854). This elevated sensitivity is particularly critical in safety-sensitive applications, where the timely detection of elevated core temperature is essential for preventing heat-related illnesses. While the improvement in AUC over certain baselines is relatively modest, the substantial reduction in RMSE is of greater practical significance. A more accurate estimation of the degree of hyperthermia allows for finer-grained monitoring and better-informed interventions, which are crucial in real-world deployment scenarios.

These results highlight the strength of RCAKF's hybrid architecture, which combines two complementary mechanisms: a learnable LSTM-based residual correction module that accounts for structured, state-dependent deviations in the observation model, and an adaptive observation noise estimator that dynamically adjusts uncertainty based on recent prediction errors. Together, these components enable RCAKF to flexibly adapt to complex physiological variations while maintaining the interpretability, stability, and real-time compatibility of classical Kalman filtering frameworks. This alignment with state-of-the-art adaptive filtering techniques, which are successfully employed in other complex domains such as lithium-ion battery management [53–55], underscores the broader relevance and timeliness of our proposed methodology.

Contextualizing the performance of our RCAKF model is important, though direct comparisons with other method types are challenging due to differences in protocols and reported metrics. Nevertheless, a qualitative assessment can be made against established benchmarks. For example, specialized non-invasive methods like zero-heat-flux (ZHF) thermometry can achieve very high precision, with reported 95 % limits of agreement of ±0.40°C against esophageal temperature in controlled exercise studies [24]. Our model's 95 % limits of agreement of ±0.61°C, while not reaching this level of precision, is a strong result for an estimation framework that relies only on a single, universally available physiological signal. The primary contribution of our approach, therefore, is not in surpassing the accuracy of hardware-intensive methods, but in offering a compelling balance of acceptable performance, convenience, and scalability for real-world, continuous monitoring where specialized equipment is impractical.

While RCAKF improves estimation performance, several limitations of this study highlight important avenues for future research.

First, the model was validated on a cohort of young, healthy adults, and its generalizability to other populations warrants discussion. Thermoregulatory capabilities, for instance, vary significantly across different populations. Children have underdeveloped cardiovascular and sweating functions, while older adults often exhibit diminished thermoregulatory responses. These age-related physiological differences would likely alter the fundamental relationship between heart rate and core temperature, requiring age-specific model calibration to ensure accuracy [56]. Similarly, individuals with underlying health conditions (e.g., cardiovascular disease) or those on medications that affect heart rate may not fit the model's assumptions. Future work should aim to validate and adapt the model on more diverse populations to broaden its applicability.

Second, the residual-compensation LSTM component was trained on data featuring relatively homogeneous activity profiles, consisting mainly of structured transitions between rest, exercise, and recovery. As such, the LSTM may partially overfit to these stereotyped temporal patterns, capturing activity transitions rather than more general physiological dynamics. Future work should therefore investigate the model's robustness using datasets that include more diverse and unstructured activity types to better assess its generalizability.

Third, the model's real-world performance is linked to the quality of the input heart rate signal. Our RCAKF, with its adaptive filtering mechanism, is inherently designed to have robustness against noisy data. Furthermore, modern commercial heart rate sensors have demonstrated substantial improvements in accuracy, allowing them to provide reasonably reliable measurements even under exercise conditions [57,58]. However, the potential impact of extreme noise events or artifacts on the model's estimation accuracy remains an area that warrants further investigation. Future work could explore the model's performance boundaries under such challenging conditions.

Finally, a more profound limitation concerns the model's specialization in hyperthermic states. The evaluation dataset exclusively captures physiological responses to exertional heat stress, meaning the RCAKF model is tuned for the dynamics of a warming body but is naive to the complexities of cooling. This is a critical point because the body's thermoregulatory responses to cold are fundamentally different from those to heat. In a hot environment or during exercise, heart rate reliably increases to meet elevated metabolic demands and to help dissipate heat. In a cold environment, however, the body employs two different primary strategies: peripheral vasoconstriction to reduce heat loss from the skin, and an increase in metabolic heat production (e.g., shivering) to generate warmth [59,60]. These distinct physiological mechanisms create a more complex relationship between heart rate and core temperature that is not captured in our training data. Therefore, a model trained

solely on heat-stress data cannot be expected to generalize accurately to hypothermic conditions. Future work should focus on developing a more robust framework, potentially a state-switching model, capable of identifying and adapting to these different physiological states to ensure accuracy across a wider range of thermal environments.

Despite these limitations, the proposed RCAKF framework offers a practical advantage by relying solely on heart rate as the input signal. This design choice provides several benefits. Heart rate is readily available from most commercial wearable devices with minimal setup and calibration [61,62]. By avoiding dependence on additional sensor modalities, RCAKF minimizes hardware complexity, computational overhead, and power consumption-making it well-suited for deployment on lightweight, low-power wearable platforms. In contrast, multimodal models that incorporate skin temperature or heat flux often require bulky or intrusive sensors, which may reduce user comfort and long-term wearability. Moreover, skin temperature is known to exhibit substantial variability depending on sensor placement and ambient conditions, and it often poorly correlates with core body temperature during exertional heat stress [11,63]. These factors introduce inconsistencies and may limit the generalizability of multimodal approaches in diverse environments.

Future work could explore several promising directions to enhance the RCAKF framework. First, the selective integration of auxiliary signals such as accelerometry, sweat rate, or body surface temperature could improve the model's context-awareness without compromising wearability. For example, recent studies on other measurement sites, like the oral cavity, highlight the importance of compensating for environmental disturbances, a task for which auxiliary sensors would be highly valuable [64]. Second, this contextual data could be used to enhance the residual correction module, allowing it to dynamically adapt based on detected activity states or environmental conditions, which would further improve performance in nonstationary settings. Finally, incorporating uncertainty estimation within the LSTM component itself would be a significant step forward, enabling the RCAKF to provide not just an estimate but also a confidence level for each prediction, which is crucial for safety-critical applications.

5. Conclusion

This study presented a Residual-Compensated Adaptive Kalman Filter (RCAKF), a novel hybrid framework for real-time core body temperature estimation using only a heart rate signal. The proposed method enhances the classical extended Kalman filter by integrating an LSTM-based residual correction module and an adaptive observation noise estimator. Experimental results on a controlled physiological dataset demonstrated the superiority of the RCAKF. It consistently outperformed five representative models across both regression and classification tasks. Specifically, the RCAKF achieved a Root Mean Squared Error (RMSE) of 0.31°C and a coefficient of determination (R^2) of 0.67, the best performance among all tested methods. It also showed the highest sensitivity in detecting elevated core temperature episodes, which is critical for safety monitoring. These findings highlight the effectiveness of combining model-based filtering with lightweight machine learning augmentation to achieve robust and deployable thermal monitoring. Future work will focus on validating the RCAKF in free-living settings and exploring its extension with uncertainty-aware decision support for personalized heat stress management.

CRediT authorship contribution statement

Yuanzhe Zhao: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Jeroen H.M. Bergmann:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Data availability statement

The implementation code for the proposed methods is available at: https://github.com/Oxford-NIL/RCAKF.git. Dataset 1 can be accessed from the Loughborough University Research Repository via: https://doi.org/10.17028/rd.lboro.26076577.v1. Dataset 2 is not publicly available due to usage restrictions but can be requested from the corresponding author of the original publication.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT 4o in order to refine sentence structure and enhance readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare no conflict of interest.

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